

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("bhanupratapbiswas/superstore-sales")

print("Path to dataset files:", path)
```

Downloading to /root/.cache/kagglehub/datasets/bhanupratapbiswas/superstore-s
100%|██████████| 478k/478k [00:00<00:00, 77.9MB/s]Extracting files...
Path to dataset files: /root/.cache/kagglehub/datasets/bhanupratapbiswas/supe

```
import os

files = os.listdir(path)
print(files)
```

```
['superstore_final_dataset (1).csv']
```

```
import pandas as pd
import os

# Create full file path
file_path = os.path.join(path, 'superstore_final_dataset (1).csv')

# Load dataset with different encoding
df = pd.read_csv(file_path, encoding='latin1')

# Show first 5 rows
df.head()
```



	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_
0	1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire
1	2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire
2	3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	Darrin Van
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Do
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Do

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
# Display dataset information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row_ID                9800 non-null   int64
1   Order_ID              9800 non-null   object
2   Order_Date            9800 non-null   object
3   Ship_Date             9800 non-null   object
4   Ship_Mode             9800 non-null   object
5   Customer_ID           9800 non-null   object
6   Customer_Name         9800 non-null   object
7   Segment               9800 non-null   object
8   Country               9800 non-null   object
9   City                  9800 non-null   object
10  State                  9800 non-null   object
11  Postal_Code           9789 non-null   float64
12  Region                 9800 non-null   object
13  Product_ID            9800 non-null   object
14  Category              9800 non-null   object
15  Sub_Category          9800 non-null   object
16  Product_Name          9800 non-null   object
17  Sales                 9800 non-null   float64
```

```
dtypes: float64(2), int64(1), object(15)
memory usage: 1.3+ MB
```

```
# Summary statistics for numerical columns
df.describe()
```

	Row_ID	Postal_Code	Sales
count	9800.000000	9789.000000	9800.000000
mean	4900.500000	55273.322403	230.769059
std	2829.160653	32041.223413	626.651875
min	1.000000	1040.000000	0.444000
25%	2450.750000	23223.000000	17.248000
50%	4900.500000	58103.000000	54.490000
75%	7350.250000	90008.000000	210.605000
max	9800.000000	99301.000000	22638.480000

```
# Mean
print("MEAN:")
print(df.mean(numeric_only=True))

# Median
print("\nMEDIAN:")
print(df.median(numeric_only=True))

# Mode
print("\nMODE:")
print(df.mode().iloc[0])
```

```
MEAN:
Row_ID      4900.500000
Postal_Code  55273.322403
Sales        230.769059
dtype: float64
```

```
MEDIAN:
Row_ID      4900.50
Postal_Code  58103.00
Sales        54.49
dtype: float64
```

```
MODE:
Row_ID      1
Order_ID    CA-2018-100111
Order_Date  5/9/2017
Ship_Date   26/09/2018
Ship_Mode   Standard Class
Customer_ID WB-21850
Customer_Name William Brown
Segment     Consumer
Country     United States
```



```
City          New York City
State         California
Postal_Code   10035.0
Region        West
Product_ID    OFF-PA-10001970
Category      Office Supplies
Sub_Category  Binders
Product_Name  Staple envelope
Sales         12.96
Name: 0, dtype: object
```

```
# Check missing values
print(df.isnull().sum())
```

```
Row_ID          0
Order_ID        0
Order_Date      0
Ship_Date       0
Ship_Mode       0
Customer_ID     0
Customer_Name   0
Segment         0
Country         0
City            0
State           0
Postal_Code     11
Region          0
Product_ID      0
Category        0
Sub_Category    0
Product_Name    0
Sales           0
dtype: int64
```

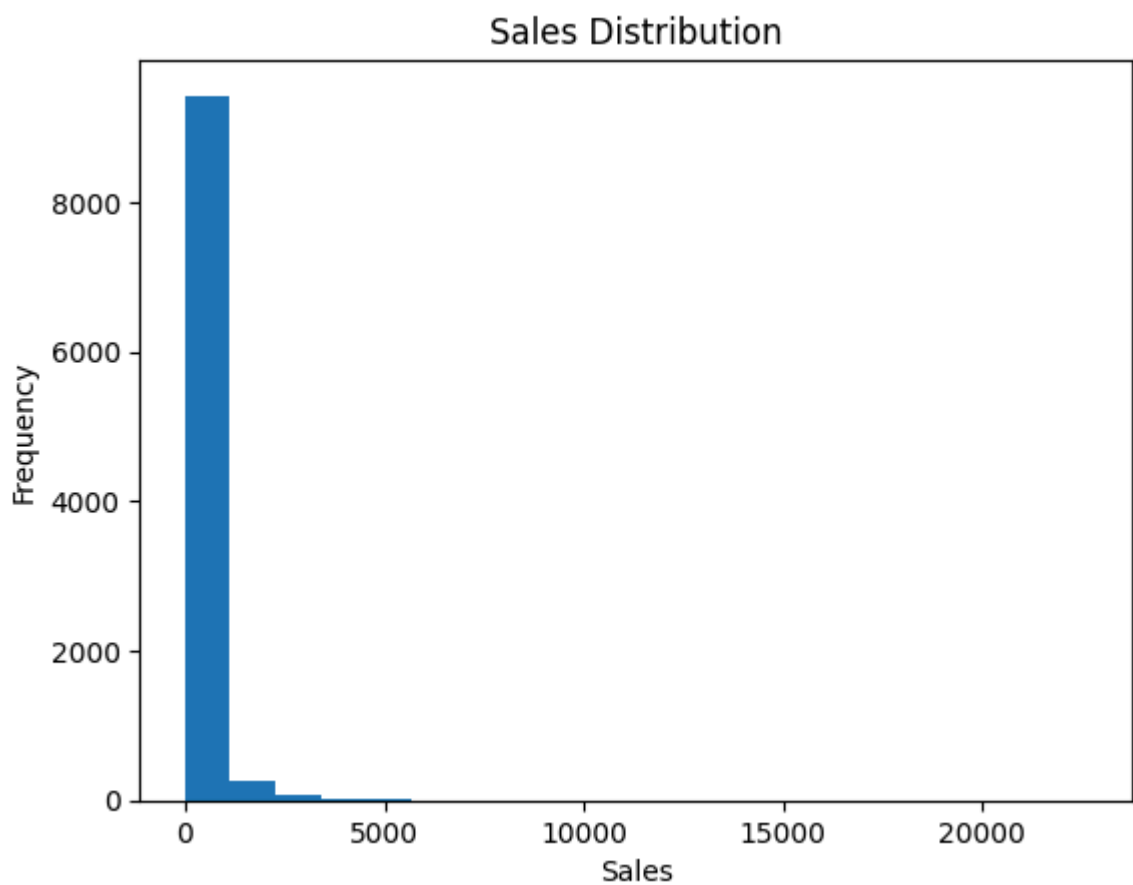
```
import matplotlib.pyplot as plt

# Histogram of Sales
plt.hist(df['Sales'], bins=20)

plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")

plt.show()
```





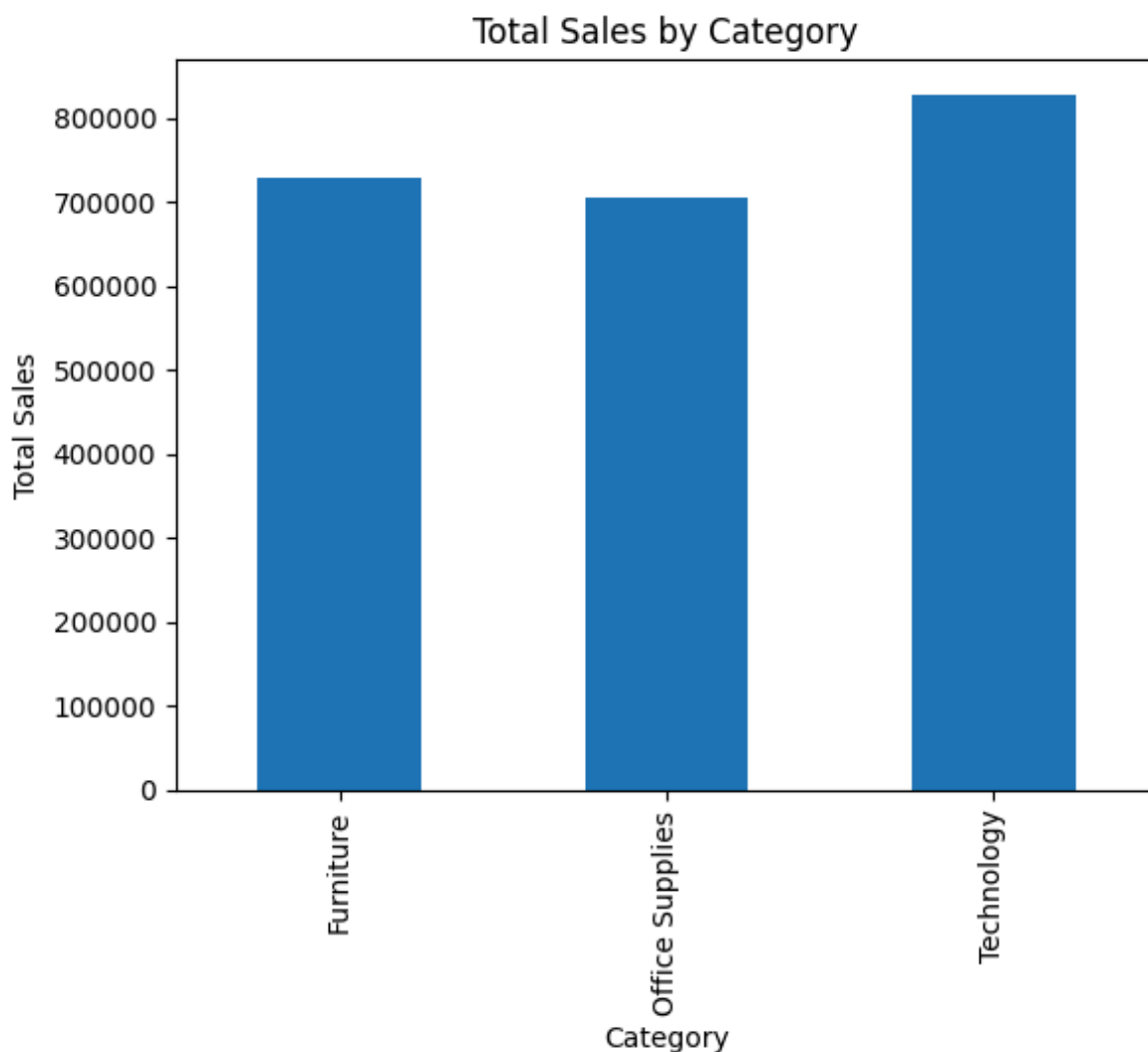
```
# Total sales by category
category_sales = df.groupby('Category')['Sales'].sum()

# Bar chart
category_sales.plot(kind='bar')

plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")

plt.show()
```





```
# Count of products by category
category_count = df['Category'].value_counts()

# Pie chart
plt.pie(category_count, labels=category_count.index, autopct='%1.1f%%')

plt.title("Category Distribution")

plt.show()
```



Category Distribution

Office Supplies

60.3%

18.5%

Technology

21.2%

Furniture

Superstore Sales Analysis – Alfido Tech

Project Overview

This project analyzes the Superstore Sales dataset using Python, Pandas, and Matplotlib in Jupyter Notebook. The objective was to understand sales patterns, identify trends, and generate useful business insights through statistical analysis and visualizations.

Tasks Performed

- Loaded and explored the dataset
- Checked data types and missing values
- Calculated summary statistics such as mean, median, and mode
- Created visualizations including histogram, bar chart, and pie chart

Key Findings

1. The sales distribution histogram showed that most sales transactions are concentrated in lower sales ranges, while only a few transactions have very high

